star’s surface, the implosion did not freeze at all. The star’s surface shrank right through the critical circumference and on inward, without hesitation.

Wheeler, in fact, had already come to expect this. Other insights (to be described below) had already transformed him from a critic of Oppenheimer’s black holes to an enthusiastic supporter. But here, for the first time, was a concrete proof from a realistic computer simulation: Implosion must produce black holes.

Was Oppenheimer pleased by Wheeler’s conversion? He showed little interest and little pleasure. At a December 1963 international conference in Dallas, Texas, on the occasion of the discovery of quasars (Chapter 9), Wheeler gave a long lecture on stellar implosion. In his lecture, he described with enthusiasm the 1939 calculations of Oppenheimer and Snyder. Oppenheimer attended the conference, but during Wheeler’s lecture he sat on a bench in the hallway chatting with friends about other things. Thirty years later, Wheeler recalls the scene with sadness in his eyes and voice.

In the late 1950s, Zel’dovich began to get bored with weapons design work. Most of the really interesting problems had been solved. In search of new challenges, he forayed, part time, into the theory of elementary particles and then into astrophysics, while keeping command of his bomb design team at the Installation and of another team that did subsidiary bomb calculations at the Institute of Applied Mathematics, in Moscow.

In his bomb design work, Zel’dovich would pummel his teams with ideas, and the team members would do calculations to see whether the ideas worked. “Zel’dovich’s sparks and his team’s gasoline” was the way Ginzburg described it. As he moved into astrophysics, Zel’dovich retained this style.

Stellar implosion was among the astrophysical problems that caught Zel’dovich’s fancy. It was obvious to him, as to Wheeler, Colgate, May, and White in America, that the tools of hydrogen bomb design were ideally suited to the mathematical simulation of imploding stars.

To puzzle out the details of realistic stellar implosion, Zel’dovich collared several young colleagues: Dmitri Nadezhin and Vladimir Imshennik at the Institute of Applied Mathematics, and Mikhail Podurets at the Installation. In a series of intense discussions, he gave them his vision of